

Project Proposal

Group Members

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Title of the Project:

Smart School Bus Tracking and Management System

Key Words:

- IOT
- IOT 3G/4G + GPS Raspberry Pi Module
- Raspberry Pi
- MANAGEMENT SYSTEM
- RFID
- APP / WEB PAGE

Scope of the Project:

This project can be implemented in the EDUCATIONAL INSTITUTES where Students/staff members travel to institute and vice versa using specified transport provided. Usually, in case of students/staff if they find problematic, waiting in the morning at the pick-up location by their respective buses or any other transport. Also, Educational Institutes are facing problems while tracking and maintaining the record of students/staff who are using buses.

With the help of this project Students/Staff, who are using the points can track their buses and may also be notified in advance if bus is near to their pick-up location. Meanwhile, for educational institutes, it helps them in maintaining the records of those who are using the Data Base Management System and can track the bus through tracking hardware installed on the buses.

Introduction:

Majority of students and staff members of educational institutes travel to their institutes from home and vice versa using Public/Institutes Transport. Many students get late and in result they miss their bus. Also, their parents are worried whether their child is on bus or not. For educational institutes, currently they have manual

data base record, it is hard for them to track the specific record if needed. Moreover, they may have no idea about the buses, if they are traveling through specified route or not.

To avoid all these problems, we have proposed the IOT Based Project using Raspberry Pi in such a way that it will be linked to Data Base Management System for storing and accessing the record, IOT 4G/4G + GPS module for tracking and RFID for ensuring the attendance. This data is sent to an app/web page for the students/staff and the Data Management System.

Project Background and Literature Review:

Wide research and development work has been accomplished on IOT and smart based school bus tracking and data management system but physically it is implemented only in developed countries. However, implementation in a country like Pakistan, this project will create new innovation to the Science and Technology in a country where no implantation is done in this modern era. Moreover, this project provides easiness to both the administration through management system and users through smart app/web-page application.

The IOT allow object to be sensed or controlled remotely across existing network infrastructure, creating opportunities for direct integration of the physical world into computer-based systems, resulting in improved efficiency, accuracy and economic benefits in addition to reduced human intervention. Where IOT is augmented class of cyber physical system which also encompasses technologies such as smart grids, virtual power plants, smart homes and smart cities. Each thing is uniquely identified through its embedded computing systems and it is also able to interoperate within the existing infrastructure.

Current State of the Art:

This project itself is the state-of-the art project in the field of technical development. It will revolutionize the older ways of transport facility available to the student/staff of any institute. Similar versions of this project have already been implemented in developed countries but in countries like Pakistan, very little or no work has been done. This project will bring dramatical changes to both industries (for making such hardware) and educational institutes to develop and use the secure and error free IOT base smart bus tracking and management system.

Challenges:

- Designing the hardware of such a system
- Designing of Data Base Management System
- Integrating the hardware, android software/web-page and Data management system
- Design an IOT Based smart system to track the shuttle position and generate an alert
- Practical Implementation of the project

Motivation and Need:

In our daily life, students travel to their institutes via Transport provided to them by respective institutes due to their safety reasons as compared to public transport. But it is very difficult for the Parents/ Institute to have proper check and balance on the provided vehicle. Thus, this project will provide help to parents in this format that a SOS message will show up on their phone when vehicle is near to their children pick-up's location, re-routing of the route to overcome the distance and time in the case if student notifies the system that he will not be using the service. Meanwhile, computerized data base record will be stored in the administration's PC.

Specific Objectives Being Addressed by the Project:

- Implementation of IOT based smart system that will automatically keep track of daily bus routine
- without human-to-human interaction
- Design and implement the system that will keep real time tracking and notifies over the system via IOT
- Integration of Hardware, PC Management and Android APP/Web page using IOT